

# Niels Slotboom

MSc Theoretical Physics / Part III Student

Location (from Oct 2025): Cambridge, UK

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## Academic Summary

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I am a theoretical physicist with a BSc and MSc in physics from the University of Bern, where I developed a strong foundation in mathematical physics, particularly general relativity, alongside growing expertise in high-performance computing. I aim to extend and refine this foundation during my studies at the University of Cambridge. My experience includes C++ and CUDA programming, with applications to numerical simulations outside gravitation. I am particularly interested in generalised theories of gravity, and aim to contribute to both their analytical study and numerical implementation—bringing together geometry, field theory, and computational methods in my future work. For more information, see the [About page](#) of my website.

## Education

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| 2025 - 2026 | <b>DAMTP, University of Cambridge</b> , Cambridge, UK<br><i>Part III, MAST in Theoretical Physics</i>                                                                                                                                                                                               |
| 2023 - 2025 | <b>Institute of Theoretical Physics, University of Bern</b> , Bern, CH<br><i>MSc in Theoretical Physics</i><br>Grade: 5.96 / 6.0, Thesis: Recovering the Eikonal Quasinormal Mode Spectrum of the Schwarzschild Black Hole via the Penrose Limit                                                    |
| 2019 - 2023 | <b>Department of Physics, University of Bern</b> , Bern, CH<br><i>BSc in Physics, Minors in Mathematics and Computer Science</i><br>Grade: 5.74 / 6.0, Thesis: Development and Theoretical Study of RF Pulses for Globally Spatially Homogeneous Excitation in Magnetic Resonance Imaging at (X)UHF |
| 2016 - 2019 | <b>Gymnasium Lerbermatt</b> , Köniz, CH<br><i>Swiss Matura</i>                                                                                                                                                                                                                                      |

## Professional Experience

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| 4/2025 - 9/2025 | <b>Bau- und Verkehrsdirektion Kanton Bern</b><br><i>Internship on Traffic Management Systems</i><br>Lead development of a LiDAR-based embedded system for vehicle detection and counting, designed for traffic light control at intersections and traffic surveys. Further, I automated data analysis processes in C++ to improve performance evaluations and reporting. |
| 2/2024 - 2/2025 | <b>Institute of Theoretical Physics, University of Bern</b><br><i>Position as Teaching Assistant</i><br>Supervision of tutorial sessions, grading exercises and exams, and answering individual questions from bachelor's students for lectures on mathematical methods in physics.                                                                                      |

## Theses

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**Master Thesis** (*Supervisor*: Matthias Blau)

*Title*: Recovering the Eikonal Quasinormal Mode Spectrum of the Schwarzschild Black Hole via the Penrose Limit

*Description*: This thesis investigated the eikonal quasinormal mode (QNM) spectrum of the Schwarzschild black hole, exploring how it emerges from the Penrose limit of the spacetime along an equatorial photon ring geodesic. The work involved three analytical approaches—WKB analysis, geometric optics, and wave propagation in the Penrose limit—all of which reproduced the known QNM spectrum. A primary result was identifying how local symmetries near the photon sphere

become exact in the Penrose limit, explaining the agreement. The project deepened my expertise in both general relativity and differential geometry. For more details, see [here](#).

#### **Bachelor Thesis** (*Supervisor*: Thomas Feurer)

*Title*: Development and Theoretical Study of RF Pulses for Globally Spatially Homogeneous Excitation in Magnetic Resonance Imaging at (X)UHF

*Description*: This thesis focused on improving parallel transmission (pTx) MRI at high field strengths, particularly at 7T, by addressing RF interference-induced excitation inhomogeneity. I developed a simulation framework based on Maxwell's equations and quaternionic Bloch's equations, implemented in CUDA C to run on the GPU. This simulation framework was embedded in a gradient descent algorithm to iteratively optimise RF pulse parameters, effectively reducing excitation inhomogeneity compared to standard circular polarisation pulses. The work demonstrates a practical approach for real-time, patient-specific pulse optimisation, addressing a critical challenge in high-field MRI.

## **Skills**

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### **Relevant Coursework**

- **Master's Courses**: General Relativity, Quantum Field Theory, Standard Model, Advanced Concepts in Theoretical Physics, Effective Field Theories, Theoretical Exercises, Thesis
- **Bachelor's Courses**: Classical Field Theories, Quantum Theory I–III, Statistical Thermodynamics I & II, Electrodynamics, Solid State Physics, Experimentalist Courses

### **Further Courses in Mathematics**

- **Mathematics**: Topology, Functional Analysis, ODEs, Non-linear PDEs, Algebra, Complex Analysis, Fourier Analysis, Real Analysis, Linear Algebra
- **Computer Science**: Programming (Java), Computer Architecture, Fundamentals of Technical Computer Science

### **Programming and Web**

- **C++ / CUDA / C**: Advanced, high proficiency in HPC and numerics, with codebases approaching senior-level design and performance
- **Java / Python**: Moderate; used for scripting, tooling, and coursework
- **HTML / CSS**: Basic; sufficient for website authoring and layout
- **x86 Assembly / PTX**: Read-level proficiency; occasionally used for performance analysis and low-level optimisation insight

### **Technical Skills**

- **Typesetting**: Typst, LaTeX
- **Hardware**: Experienced in Computer Configuration and Assembly, Hardware Architecture, Embedded Systems
- **Electronics**: Circuit and PCB Design, Repairs, Reverse Engineering
- **General**: Troubleshooting, Networking

### **Complementary Skills**

- **Teaching**: Experienced in mentoring peers and serving as a teaching assistant, with a focus on adapting explanations to individual learning needs

### **Languages**

- **German, Dutch**: Native
- **English**: C2, CEFR 209, acquired in 2024
- **French**: Baccalaureate level